How to Prepare For ML/DS Interviews

Journey

Choose the role -> Build the evidence -> Find opportunities -> Prepare -> Practice -> Interviews -> Review Feedback -> Improve

1. **Choose the work/role -**
    a. Data Scientist - Stats, Experimentation, SQL (optional), Modelling, Business Interpretation, Coding
    b. ML Engineer - Coding, ML Pipelines, Deployment, Monitoring
    c. Applied/Research Scientist - Mathematical depth, modelling experiments, research papers (adapting, writing), novel approaches
    d. GenAI Engineer - LLMs, retrieval, evaluation, inference, application engg, prompt engg.
    e. MLOps Engineer - Infra, CI/CD pipeline, platform reliability and monitoring
    f. GenAI Architect - LLM Finetuning
    g. Forward Deployed Engineer - Understanding business and designing solution
2. **Interview Loop:**
    a. Recruiter Screening
    b. Hiring Manager
    c. Coding/SQL - DSA, ML Coding round
    d. ML Breadth - Breadth questions from ML/DL
    e. Project Depth - Dive deep into one of the projects
    f. ML System Design - Given a problem statement, design a full ML pipeline
    g. Behavioral/Leadership - Given a scenario, how did you responds
3. **Role Hypothesis:**
    a. Collect min 5 JDs
    b. Underline repeated responsibilities
    c. Separate the requirements

d. Predict the interview

4. **ML Revision**
    a. Recall - Explain the topics from memory before opening a resource.
    b. Repair - study the missing/incorrect part
    c. Apply - Use it on an example or small calculation
    d. Speak - Give a concise loud answer, or write it up
    e. Links:
        i. https://developers.google.com/machine-learning/crash-course
        ii. https://cs229.stanford.edu/, CS231N
        iii. https://developers.google.com/machine-learning/guides/rules-of-ml/

    What the grind actually looks like:
        1. Motivation
        2. Friction
        3. Exposure
        4. Repair
        5. Recognition
5. How to Answer:
    a. Clarify the context
    b. Define it directly and then expand it
    c. Build the intuition in ordinary language
    d. Explain the Math
    e. State the assumption
    f. Compare the trade offs
    g. Apply it on an example
    h. How to handle follow up questions:
        i. Dont defend instantly.
        ii. Tell your assumptions
        iii. Always look for other relevant answers
6. ML System Design:

a. Define the problem statement
b. Define the constraints
    i. Volume
    ii. Latency
    iii. Explainable
    iv. Geographic
    v. Cost
    vi. Scalability
c. Define the data and labels
d. Establish a baseline
e. Metrics
    i. Connect with business outcomes
    ii. Offline metrics and online metrics
f. Improve the modelling
    i. Try different algorithms
    ii. Different features
    iii. Generalization technique, hyperparameter tuning
    iv. Error analysis
g. Monitoring
    i. Online metrics
    ii. Data drift
    iii. Prediction Quality
    iv. Analysis of the prediction
h. Re-training/Finetuning

7. Profile and Resume
    a. For LinkedIn
        i. Improve headline: simple language, target role, evidence, specialization, unordered (structured) list of tools you know
        ii. About - Present focus, 2/3 proof points, direction
        iii. Experience - Problems, ownership, decisions, outcome
        iv. Featured- Resume, papers, media links, demo, repository
        v. Skills
        vi. Settings - work mode, contact details, job preferences

vii. Hack for linkedin - On a weekly (2-3 days are preferred) basis, keep updating your profile (for the sake of update), your profile comes in a good amount of searches.

b. Resume
    i. Template
        1. Overleaf - https://www.overleaf.com/gallery/tagged/cv
        2. Word templates - https://create.microsoft.com/en-us/templates/resumes
    ii. Pattern for experiences - Built/improved [system] using [method], achieving [result on metric] on [scale], leading to [outcome].
    iii. Update the resume based on the JD
        1. Real time on Responsibilities with your matching experience
        2. Long term on requirements

8. Jobs
    a. Roles
        i. Primary role - define it clearly
        ii. Adjacent roles - different title, overlapping evidence/work mostly
    b. Problems: What problems I have solved?
    c. Methods:
    d. Platforms:
        i. LinkedIn - Established Industries, not good for startup
        ii. Naukri/Indeed
        iii. YCombinator - Good for startups
        iv. InstaHyre
        v. HiringCafe
        vi. Wellfound
        vii. WelcomeToTheJungle
        viii. Himalayas

        ix. Weekday
        x. Careers portal from different companies
    e. JD Understanding
        i. Outcome - Predict, automate, Improve
        ii. Day to day responsibility - build, analyse, deploy, maintain
        iii. Technical depth required
        iv. Interview Prediction: Try to come up with questions/projects that company can ask for
        v. Resume update from this JD
9. How to Apply/Outreach
    a. JD handy, job link
    b. For role mentioned in JD, search for people on linkedin with that role
    c. Just message them (2-3 people) with a 300 characters message with job id and job link with a polite request on can you please refer me to the recruiter/hiring manager.
    d. If you are a connection - write a para on what you are working on, job link and job id, resume and a request
10. Mock Interviews
    a. You can ask your peers to have interviews based on the JD
    b. LinkedIn, request someone to take a mock interview or how to prepare
    c. Use voice mode GPT/claude to conduct an interview, set a prompt on the level of the interview and ask it to be honest in evaluations.